

Neural Cellular Automata (NCA) Evaluation Metrics

This document explains **each evaluation metric used in the NCA evaluation code**, what it measures, how it is computed, and **why it is necessary specifically for NCAs** (and not just generic image models).

The key idea: **NCAs are dynamical systems, not one-shot generators**. Therefore, evaluation must capture **accuracy, dynamics, stability, robustness, and efficiency**.

1. Final Reconstruction Error (FRE)

What it measures

How close the *best generated state* is to the target image.

This answers:

Did the NCA ever reach the correct attractor?

How it is computed

- Track the timestep with the **highest PSNR** (best state)
- Compute Mean Squared Error (MSE) between that state and the target

$$\text{FRE} = \|x_{\text{best}} - x^*\|_2^2$$

Why it is needed

NCAs can: - Overshoot - Oscillate - Degrade after a good solution

Final-step error is therefore misleading. FRE correctly evaluates the **best attractor reached**.

Interpretation

- Low FRE → correct attractor learned
 - High FRE → wrong or blurry attractor
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2. Best-State PSNR (PSNR*)

What it measures

Peak visual reconstruction quality achieved during growth.

How it is computed

$$\text{PSNR}^* = -10 \log_{10}(\text{FRE})$$

Computed at the timestep where PSNR is maximized.

Why it is needed

Standard PSNR assumes: - One-shot inference - Monotonic convergence

NCA's violate both assumptions.

Interpretation

- <15 dB: incorrect / blurry structure
 - 20–30 dB: recognizable but imperfect
 - 30 dB: high-fidelity reconstruction
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3. Time-to- ϵ Convergence (T ϵ C)

What it measures

Speed of self-organization.

Answers:

How quickly does global structure emerge?

How it is computed

For a threshold ϵ (e.g. 30 dB):

$$\text{T}\epsilon\text{C} = \min_t \{\text{PSNR}_t \geq \epsilon\}$$

Why it is needed

Two NCA's may reach the same quality but: - One takes 300 steps - Another takes 3000 steps

Standard metrics treat them equally. Dynamical systems should not.

Interpretation

- Lower T ϵ C $\rightarrow$ faster information propagation
 - Higher T ϵ C $\rightarrow$ slow, diffusion-limited dynamics
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4. Convergence Stability Index (CSI)

What it measures

How stable the system is **after convergence**.

Answers:

Does the NCA settle or keep fluctuating?

How it is computed

Variance of PSNR over a window after the best timestep:

$$\text{CSI} = \text{Var}(\text{PSNR}_{t:t+K})$$

Why it is needed

NCA's continue evolving even after convergence. Instability indicates: - Oscillatory attractors - Chaotic dynamics

Interpretation

- Low CSI → stable fixed point
 - High CSI → unstable or oscillatory behavior
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5. Steady-State Drift (SSD)

What it measures

Whether the attractor is a true fixed point of the dynamics.

Answers:

If I keep running the NCA, does it stay the same?

How it is computed

After convergence, run the model for additional steps:

$$\text{SSD} = \frac{1}{K} \sum \|x_{t+1} - x_t\|^2$$

Why it is needed

Feed-forward models do not evolve over time. NCA's do.

SSD directly measures **dynamical stability**, not visual quality.

Interpretation

- $SSD \approx 0 \rightarrow$ fixed-point attractor
 - Moderate SSD $\rightarrow$ limit cycle
 - High SSD $\rightarrow$ chaotic system
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6. Basin Robustness Score (BRS)

What it measures

Robustness of the attractor to perturbations.

Answers:

Does the system self-heal?

How it is computed

1. Take best state
2. Apply perturbation:
3. Gaussian noise
4. Spatial erasure
5. Resume evolution
6. Measure recovered PSNR

Final BRS is the average recovered PSNR.

Why it is needed

Self-repair is a **core claim of NCA models**. Pixel accuracy alone does not capture this.

Interpretation

- High BRS $\rightarrow$ wide basin of attraction
 - Low BRS $\rightarrow$ fragile solution
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7. Parameter Efficiency Ratio (PER)

What it measures

Quality achieved per parameter.

How it is computed

$$PER = \frac{PSNR^*}{\log(\#params)}$$

Why it is needed

NCAAs are about **local rules**, not brute-force capacity.

Log scaling avoids unfairly penalizing slightly larger models.

Interpretation

- High PER → compact, expressive rule set
 - Low PER → bloated architecture
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8. Time-Normalized Quality (TNQ)

What it measures

Quality achieved per unit training time.

How it is computed

$$\text{TNQ} = \frac{\text{PSNR}^*}{\text{training time}}$$

Why it is needed

NCAAs are iterative and expensive to train. Speed matters.

Interpretation

- High TNQ → efficient convergence
 - Low TNQ → computationally expensive dynamics
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9. NCA Efficiency Score (NES) — *Novel Metric*

What it measures

Overall quality of a learned self-organizing rule.

Combines: - Accuracy - Speed - Model size

How it is computed

$$\text{NES} = \frac{\text{PSNR}^*}{\# \text{params} \times \text{T\&C}}$$

Why it is important

This metric: - Penalizes slow convergence - Penalizes large models - Rewards fast, compact self-organization

There is no equivalent in GAN or diffusion evaluation.

Interpretation

- High NES → strong self-organizing dynamics
 - Low NES → inefficient rule
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10. Hidden-State Energy Ratio (HSER)

What it measures

How much internal computation the NCA uses compared to visible output.

How it is computed

$$\text{HSER} = \frac{E[\|h\|^2]}{E[\|rgb\|^2]}$$

Why it is needed

Distinguishes: - True NCAs with memory and coordination - Shallow CNN-like behavior

Interpretation

- High HSER → memory-rich, coordinated dynamics
 - Low HSER → shallow, feed-forward-like behavior
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Big Picture Summary

These metrics jointly evaluate NCAs as **dynamical systems**, not static generators. They measure: - Accuracy (FRE, PSNR*) - Emergence speed (T_εC) - Stability (CSI, SSD) - Robustness (BRS) - Efficiency (PER, TNQ, NES) - Internal dynamics (HSER)

Together, they provide a **holistic and principled evaluation framework** that standard image-generation metrics cannot offer.